Math 1200 Lecture 6

Sep 22 2026

Topics: 14.5: Chain rule, implicit differentiation Name:_____
Collaborators:_____

Group work 1 A Canadian goose is migrating south during the winter.

(a) While it is flying, for every meter of elevation it gains, the temperature T of the air drops at a rate 0.1 $°C/m$. If it is gaining elevation at 10 m/hr, how fast it its temperature changing? What is $\frac{dT}{dt}$ in $°C/hr$?

(b) For every kilometer the goose travels south, the temperature *rises* at a rate of 0.01 $°C/km$. If the goose is gaining elevation at a rate of 10 m/hr and traveling south at 50 km/hr, how fast is its temperature changing in $°C/hr$?

Group work 2 Consider the parametric curve $\vec{r}(t) = \langle x(t), y(t) \rangle = \langle t\cos(t), t\sin(t) \rangle$ and the function $f(x, y) = x^2 + y^2$. Find $\frac{df}{dt}$.

Group work 3 (a) Consider the ellipsoid $x^2 + \frac{y^2}{2} + \frac{z^2}{3} = 1$. Calculate $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the point $(x, y, z) = (1/2, 0, 3/2)$.

(b) Consider the equation $x^2 + \sin(5z) = y^3 + z$. Find $\frac{\partial y}{\partial x}$ at the point $(x, y, z) = (1, 1, 0)$.

Scan to see a graph